

Empirical Method in Finance

With

Python

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2026

Introduction

Basic models in finance:

- The **Markowitz's** approach to portfolio management.

Let μ_i and σ_i be the expected return and volatility associated to asset i . In the single asset case, the investor should invest its money accordingly to the Sharpe ratio multiplied by risk aversion:

$$\omega_i = \frac{\mu_i}{\sigma_i} \times \frac{1}{\gamma - 1}, \quad (1)$$

where γ is the relative risk aversion of the representative agent.

- The **Capital Asset Pricing Model (CAPM)**:

$$r_i = r_f + \frac{\sigma_{i,m}}{\sigma_m^2} (r_m - r_f) \quad (2)$$

Introduction

- The **Vasicek** term structure model:

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t \quad (3)$$

- The **Black Scholes** model:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad (4)$$

- ... and the risk management approach of many banks: VaR computation from Monte Carlo scenario involving both expectation and volatility, the tracking errors essential to fund management...

These models are based on the assumption that *we can measure expected return and volatility*. This is however in general false: we need an empirical strategy.

What is estimation?

Defining estimation:

"In statistics, any of numerous procedures used to calculate the value of some property of a population from observations of a sample drawn from the population."

Source: Encyclopedia Britannica.

"In statistics, an estimator is a rule for calculating an estimate of a given quantity based on observed data: thus the rule (the estimator), the quantity of interest (the estimand) and its result (the estimate) are distinguished."

Mosteller, F.; Tukey, J. W. (1987) [1968]. "Data Analysis, including Statistics". The Collected Works of John W. Tukey: Philosophy and Principles of Data Analysis 1965–1986. 4. CRC Press. pp. 601–720 [p. 633].

The aim of statistics

⇒ Random variables have characteristics that statistics aims at exploring. It requires:

- A data sample: a set of realizations for a given phenomenon.
- A quantity we want to explore: a moment of a supposed distribution, the parameters of a fitting distribution, a goodness of fit of a model, the confidence of an estimation...
- A consistent and possible efficient estimation strategy.

This class is about **models and estimation strategies** .

Outline of the Class

1. Characteristics of Financial Time Series
2. Estimation Strategies of the Linear Model
3. Time Series Analysis: Univariate Analyses
4. Time Series Analysis: Vector Autoregressive Models
5. Non-Stationarity and Cointegration
6. Gaussian Garch Models
7. Modeling Non-normality
8. Empirical Option Pricing
9. Modeling Correlation
10. The Quant Toolbox: PCA and Non-Parametric Estimates

Evaluation

Evaluation based on

- A series of compulsory group assignments, following every three classes (should be about 4 of them).
- A final exam, with closed book. Format to be specified around the last two sessions, surveying your knowledge of the whole class.

Final Grade = 60% Assignments + 40% Final Exam.

Readings

Core readings:

- Singleton, K. J. (2006). Empirical Dynamic Asset Pricing: Model Specification and Econometric Assessment. Princeton University Press.
- Jondeau, E., Poon, S., and Rockinger, M. (2006). Financial Modeling Under Non- Gaussian Distributions. Springer Finance Textbook.
- Ielpo, F., Merhy, C., & Simon, G. (2017). Engineering Investment Process: Making Value Creation Repeatable. Elsevier. See chapters 1-3.

Other readings:

- J. Hull. Options, futures and other derivatives. Prentice Hall.
- Ch. Gouriéroux and J. Jasiak. Financial Econometrics. Princeton Press.

Class 1: Exploring Returns

Outline of Class 1

- Computing Asset Returns
- Expectations of Returns
- Volatility's Stylized Facts

Simple return

Two main reasons for investigating the properties of returns instead of the properties of prices:

- Investors are mostly interested in returns for their investment decisions;
- Properties of returns are in general easier to handle than the properties of price.

Holding the asset for one period from $t - 1$ to t yields the **one-period simple return**.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \text{ or } P_t = P_{t-1} (1 + R_t)$$

where

P_t : price of the asset at date t ,

R_t : one-period simple return from date $t - 1$ to date t .

Continuously compounded (or log-) return

If we want to pay an annual interest of $R_t^{(m)}$, split into m payments in a year, the interest rate for each payment is $R_t^{(m)}/m$ and the net value of the deposit one year later will be $(1 + R_t^{(m)}/m)^m$.

In the limit case where the interest is paid continuously ($m \rightarrow \infty$), we have

$$\lim_{m \rightarrow \infty} \left(1 + \frac{R_t^{(m)}}{m} \right)^m = \exp(r_t)$$

with $P_t = P_{t-1} \times \exp(r_t)$

The **continuously compounded (or log-) return** r_t is given by

$$r_t = \log \frac{P_t}{P_{t-1}}$$

Temporal aggregation (multiple-period return)

Holding the asset for **k periods** from t to $t + k$ yields the **k -period simple return**:

$$R_t[k] = \frac{P_{t+k} - P_t}{P_t}$$

or

$$P_{t+k} = P_t (1 + R_t[k]) = P_t (1 + R_{t+1}) \times \dots \times (1 + R_{t+k})$$

so that

$$R_t[k] = \left(\prod_{j=1}^k (1 + R_{t+j}) \right) - 1$$

If one uses log-returns, the **k -period log return** is simply the sum of continuously compounded one-period returns, so that

$$r_t[k] = \log (1 + R_t[k]) = \log \left(\prod_{j=1}^k (1 + R_{t+j}) \right) = \sum_{j=1}^k \log (1 + R_{t+j})$$

so that $r_t[k] = \sum_{j=1}^k r_{t+j}$.

Contemporaneous aggregation (Portfolio return)

Continuous compounding has a drawback, when computing the return of a portfolio.

The simple return of an N -asset portfolio is the weighted average of the simple returns of the N assets. Let p be the portfolio with weight w_i on the asset i .

The **simple portfolio return** $R_{p,t}$ is

$$R_{p,t} = \sum_{i=1}^N w_i R_{i,t}$$

The **log portfolio return** $r_{p,t}$ is

$$r_{p,t} = \log \left(\sum_{i=1}^N w_i e^{r_{i,t}} \right) \neq \sum_{i=1}^N w_i r_{i,t}$$

Using simple returns or log-returns is an empirical issue, which depends in general on the problem at hand.

Other definitions

Remark 1: Dividend payments

For assets with periodic dividend payments, asset returns have to be redefined.

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1 \text{ and } r_t = \log(P_t + D_t) - \log P_{t-1}$$

where D_t is the dividend payment of an asset between dates $t - 1$ and t .

P_t is the price of the asset at the end of period t (dividend not included in P_t). That is the difference between "price return" and "total return".

Remark 2: Excess returns

Excess return is simply the difference between the asset's return and the return on the risk-free asset, in practice the short-term Treasury bill return.

$$Z_{i,t} = R_{i,t} - R_{f,t} \text{ and } z_{i,t} = r_{i,t} - r_{f,t}$$

with $R_{f,t}$ and $r_{f,t}$ the simple return and log return of the risk-free asset.

Today's Dataset

A data set of 7 different assets/indices covering the 1999-2021 period:

- MSCI World Equities
- MSCI Emerging Equities
- World Government Bonds
- High Yield Bonds Index (US)
- Oil (WTI)
- The US Dollar Index
- Gold

First Steps in Python

```
import os
os.chdir("//fileserver2//PersonalFolders$//fie//My Documents//")
print("Current working directory: {0}".format(os.getcwd()))

# Loading the data
import pandas
df = pandas.read_csv("Data_HEC_1.csv", sep = ";", decimal = ".")
dates=df.iloc[:,0]
df = df.iloc[ : , 1:8]

# Simple operations
import numpy as np
formattedData=np.array(df)
num_lines=np.size(formattedData,0)
num_col=np.size(formattedData,1)
logPrice=np.log(formattedData)
logReturns=logPrice[2:num_lines,:]-logPrice[1:(num_lines-1),:]
simpleReturns=np.divide(formattedData[2:(num_lines),:],
formattedData[1:(num_lines-1),:])-1
```

First Steps in Python

```
# Formating dates in Python
import pandas as pd
datesFormatted=pd.to_datetime(dates, format="%d.%m.%Y")

# Plot
import matplotlib.pyplot as plt

# plot of prices
plt.plot(datesFormatted,formattedData[:,0])
plt.ylabel('World Equities')
plt.show()

# Same version but with red dots
plt.plot(datesFormatted,formattedData[:,0], 'ro')
plt.ylabel('World Equities')
plt.show()
```

First Steps in Python

```
# Log plot of all series
plt.plot(datesFormatted, logPrice)

# Log plot of the first two series with different sizes
plt.plot(datesFormatted, logPrice[:, 0], 'r--', datesFormatted,
logPrice[:, 1], 'b--')

# Multiplots
plt.subplot(221)
plt.plot(datesFormatted, logPrice[:, 0], 'r--')
plt.ylabel('World Equities')
plt.subplot(222)
plt.plot(datesFormatted, logPrice[:, 1], 'r--')
plt.ylabel('Emerging Equities')
plt.subplot(223)
plt.plot(datesFormatted, logPrice[:, 2], 'r--')
plt.ylabel('World Govt Bonds')
plt.subplot(224)
plt.plot(datesFormatted, logPrice[:, 3], 'r--')
plt.ylabel('US High Yield')
```

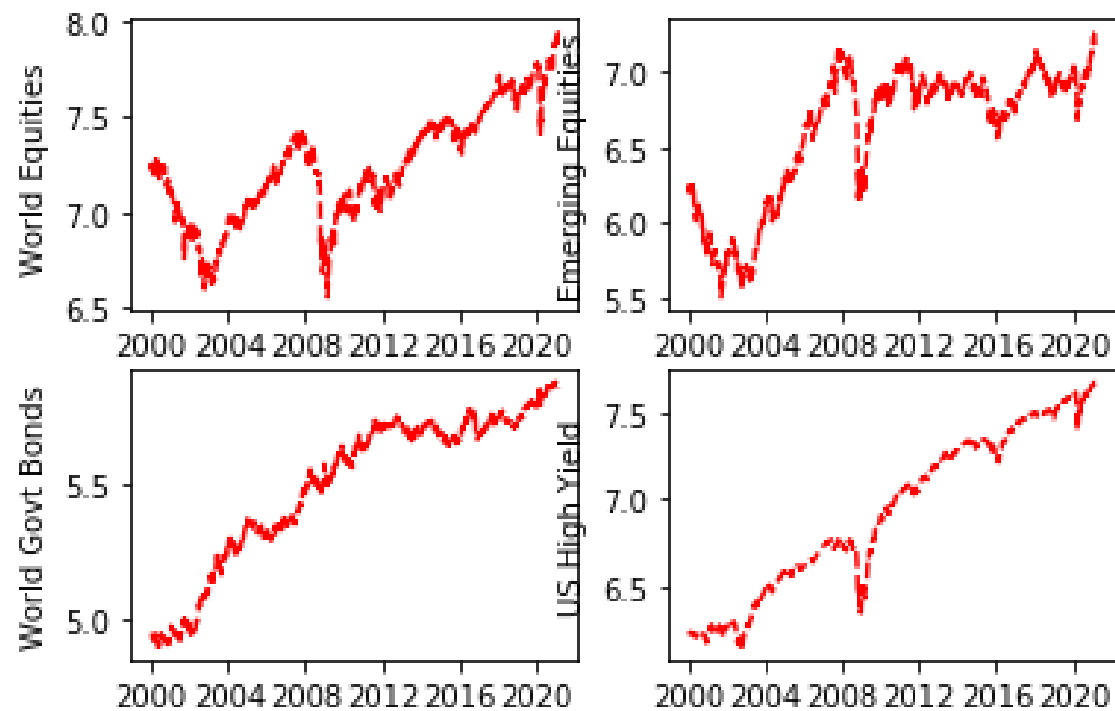
First Steps in Python

```
# Comparison of Plot Returns
plt.subplot(321)
plt.plot(datesFormatted[2:num_lines], logReturns[:,0])
plt.title('World Equities')
plt.subplot(322)
plt.plot(datesFormatted[2:num_lines], logReturns[:,1])
plt.title('Emerging Equities')
plt.subplot(325)
plt.plot(datesFormatted[2:num_lines], logReturns[:,2])
plt.title('World Govt Bonds')
plt.subplot(326)
plt.plot(datesFormatted[2:num_lines], logReturns[:,3])
plt.title('US High Yield')
```

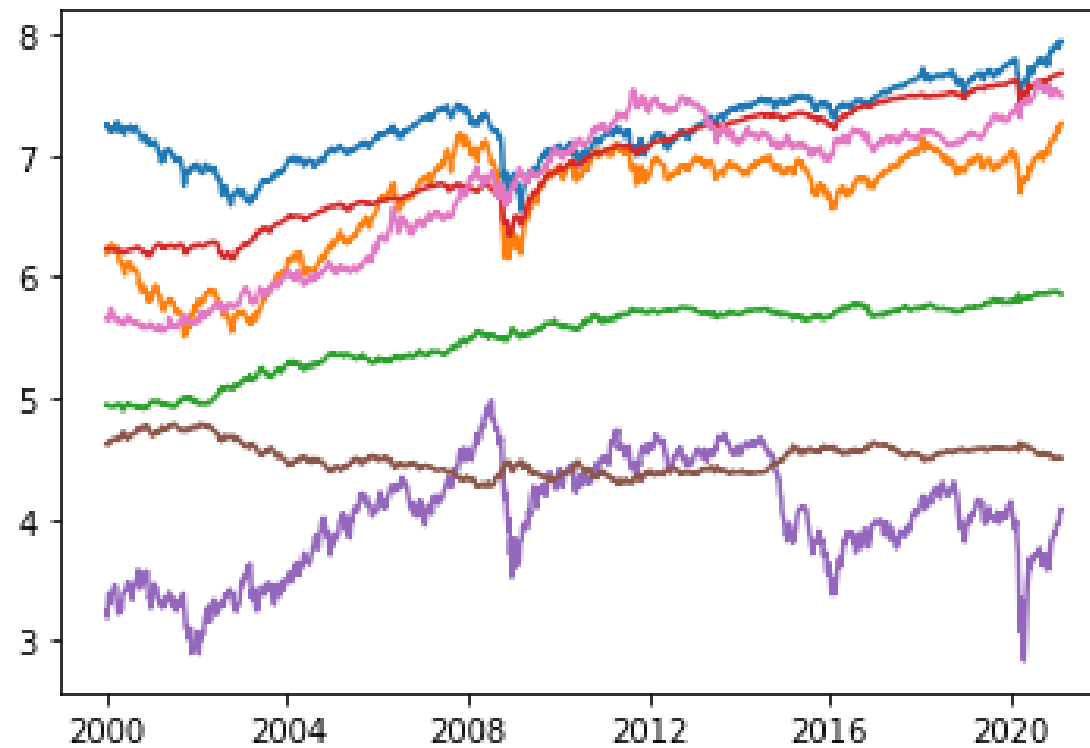
World Equities



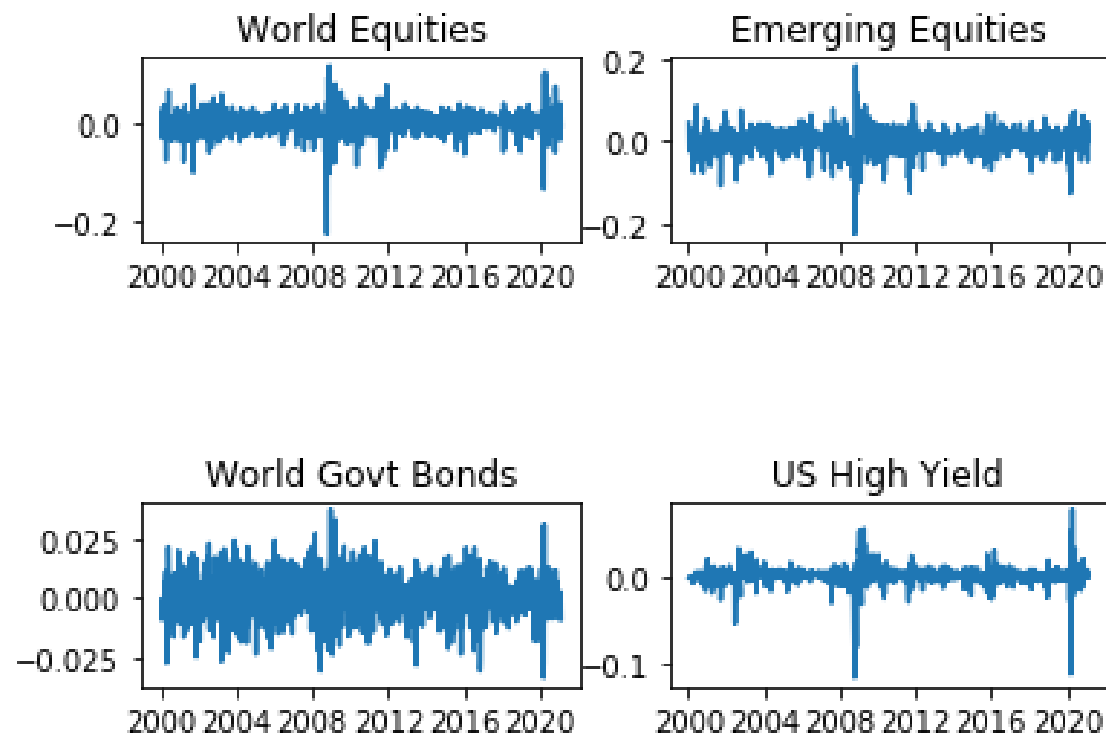
Asset Log Prices



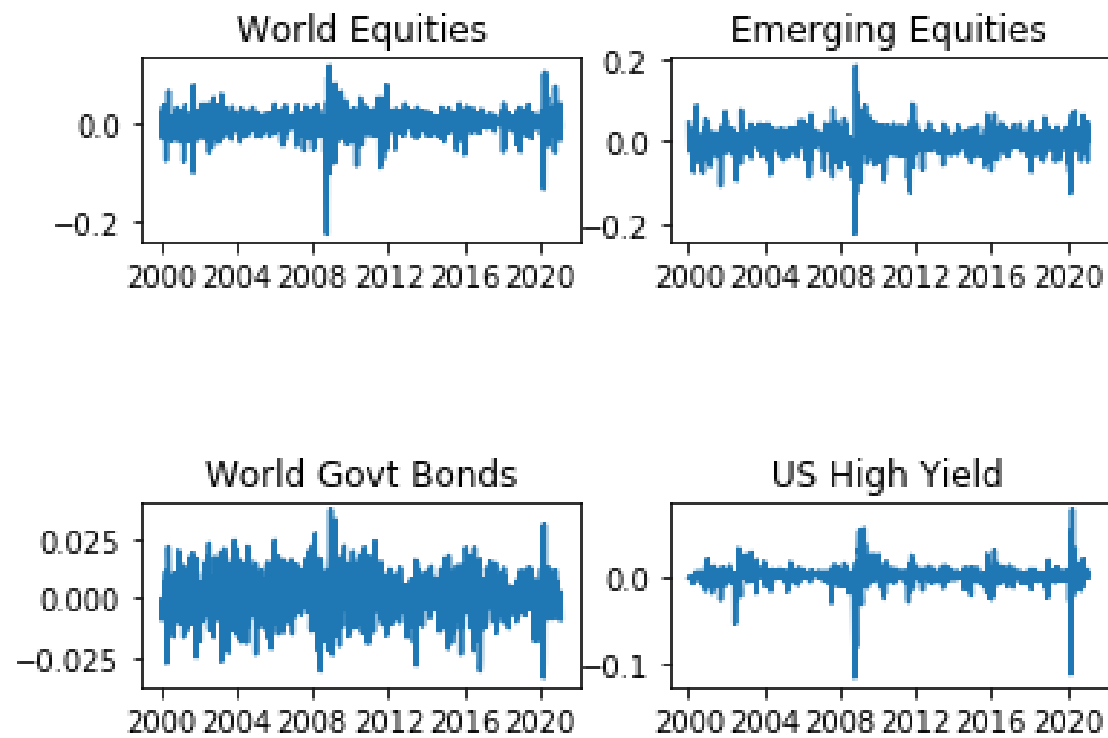
Asset Log Prices



Asset Log Prices



Asset Returns



Characteristics of Financial Asset Returns

Early work in Finance has supposed some very strong assumptions for asset returns. The most crucial assumptions are

1. **Normality of log-returns**

It is a convenient assumption for many applications in Finance (cf. Black-Scholes model for option pricing).

For stock-index returns, it is consistent with the Law of Large Numbers.

2. **Time independence (i.i.d. process)**

It is, to some extent, an implication of the Efficient Market Hypothesis. Actually, the EMH only imposes unpredictability of returns.

Moments of a random variable

Assume X (log-returns) has the following **cumulative distribution function** (cdf):

$$F_X(x) = \Pr[X \leq x] = \int_{-\infty}^x f_X(u) du.$$

where $f_X(x)$ is the **probability distribution function** (pdf).

The **mean** or expected value of X is:

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

The **variance** of X is:

$$\sigma^2 = V[X] = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) dx$$

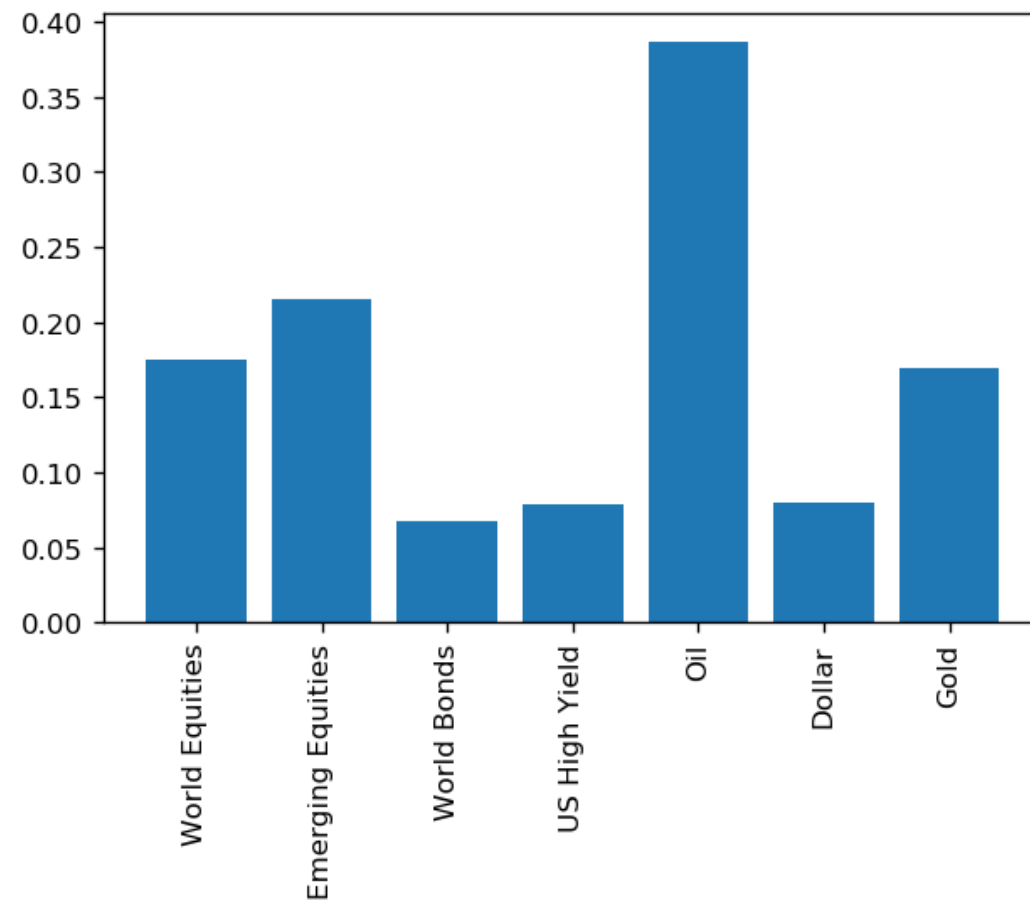
AKA "volatility".

Computing Volatility in Python

```
# non-centered moment of order 2
SquaredRet=np.power(simpleReturns, 2);
VarianceRet=np.mean(SquaredRet, 0)-np.power(np.mean(simpleReturns, 0),2);
VolatilityRet=np.power(VarianceRet,0.5);
VolatilityRet=VolatilityRet*np.power(52,0.5);
print(VolatilityRet)
```

```
labels=list(df);
plt.figure(dpi=120)
plt.bar(labels[0:8],VolatilityRet)
plt.xticks(rotation=90)
```

Asset Returns' Volatility



Scales of volatility

When facing anything new, rough estimates are essential. Volatility is always presented in terms of annualized volatility.

Suppose $(r)_{t \in \mathbb{Z}}$ is the daily returns stationary process, with

$$\mathbb{V}[r_t] = \sigma^2 \quad \mathbb{E}[r_t r_{t-1}] = 0 \quad (5)$$

where σ is the daily volatility. The annualized variance is then

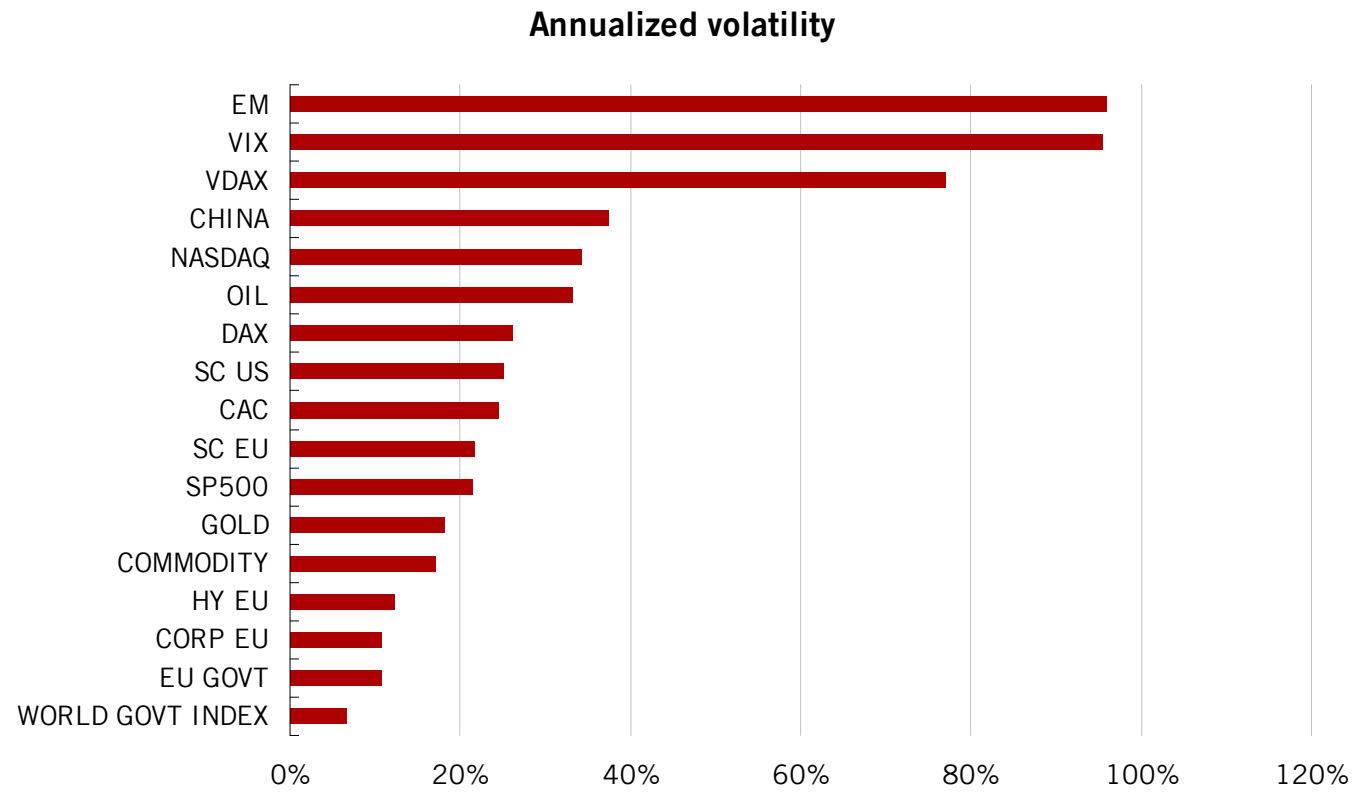
$$\mathbb{V} \left[\sum_{t=1}^T r_t \right] = \sum_{t=1}^T \mathbb{V}[r_t] = T \times \sigma^2 \quad (6)$$

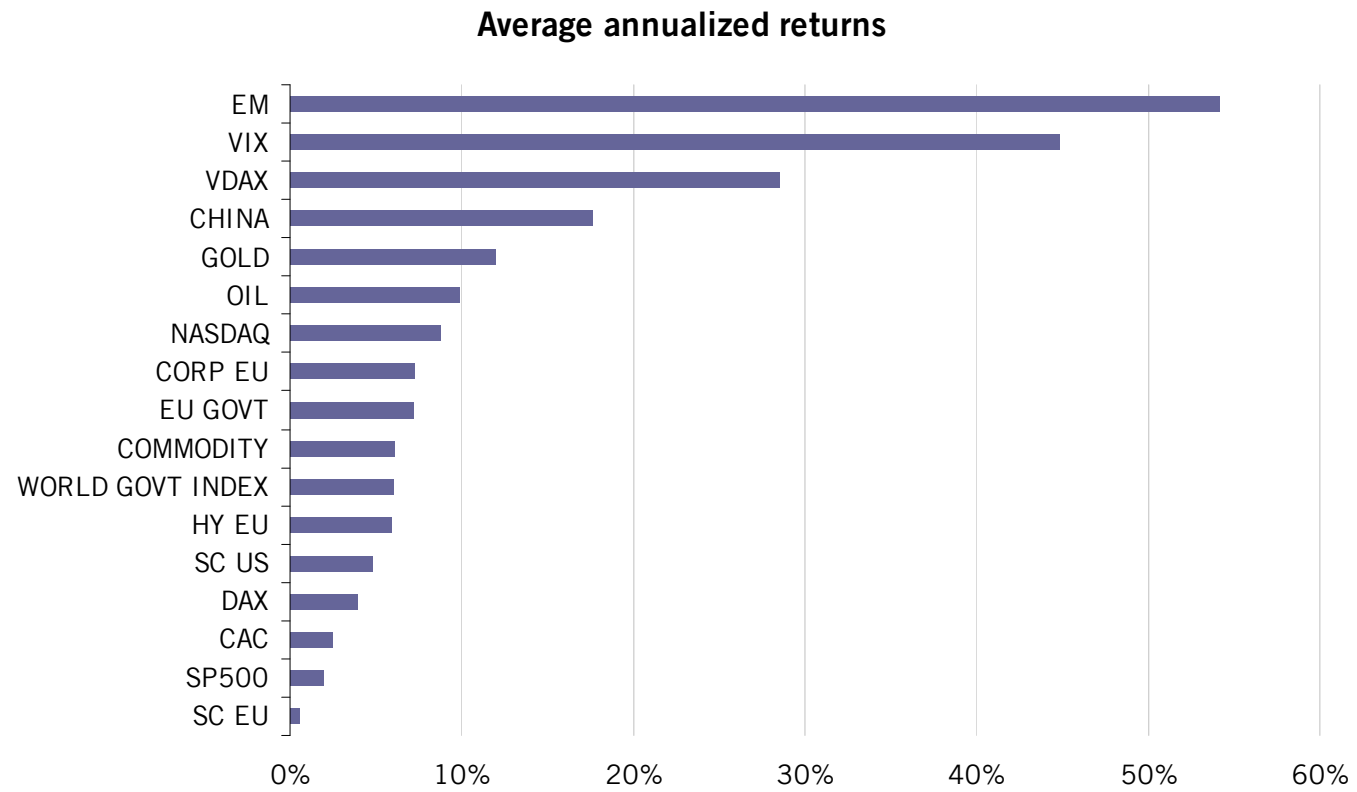
Hence, the rule of the thumb: annualized volatility is equal to

- $\sqrt{252} \times$ the daily volatility
- $\sqrt{52} \times$ the weekly volatility
- $\sqrt{12} \times$ the monthly volatility

In general, the correlation is non zero and the annualized vol from daily data is higher than the annualized vol computed from monthly data.

The lower the sample frequency and the more Gaussian the data.





Economic fundamentals to volatility

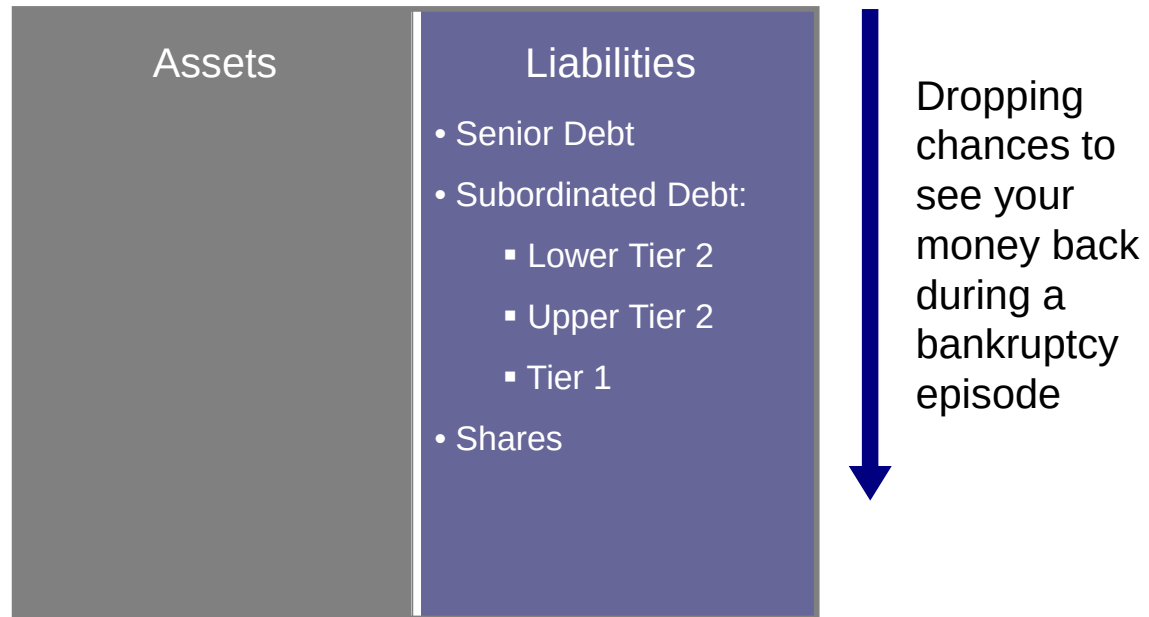
Volatility arises during periods of fear and uncertainty. Differences in volatility is related to the exposure to risk factors.

When it comes to Bonds and Stocks, the difference in volatility stems from different factors:

- Debt holder and share holder do not bear the same level of risk: during bankruptcy, debt is payed back before shares.
- The situation is the same amongst of bond holder: the degree of seniority leads to different levels of risk.
- The industry type as well as the rating of the firm explains the different in risks, from investment to speculative grade/high yield.
- Finally, the country to which the firm is mostly related can bear different risks, such as a political risk.

In the case of commodities, supply and demand games explain the hierarchy of risks.

Balance Sheet and Risk



Stylized facts #1: volatility is time varying

Risk perception changes through time.

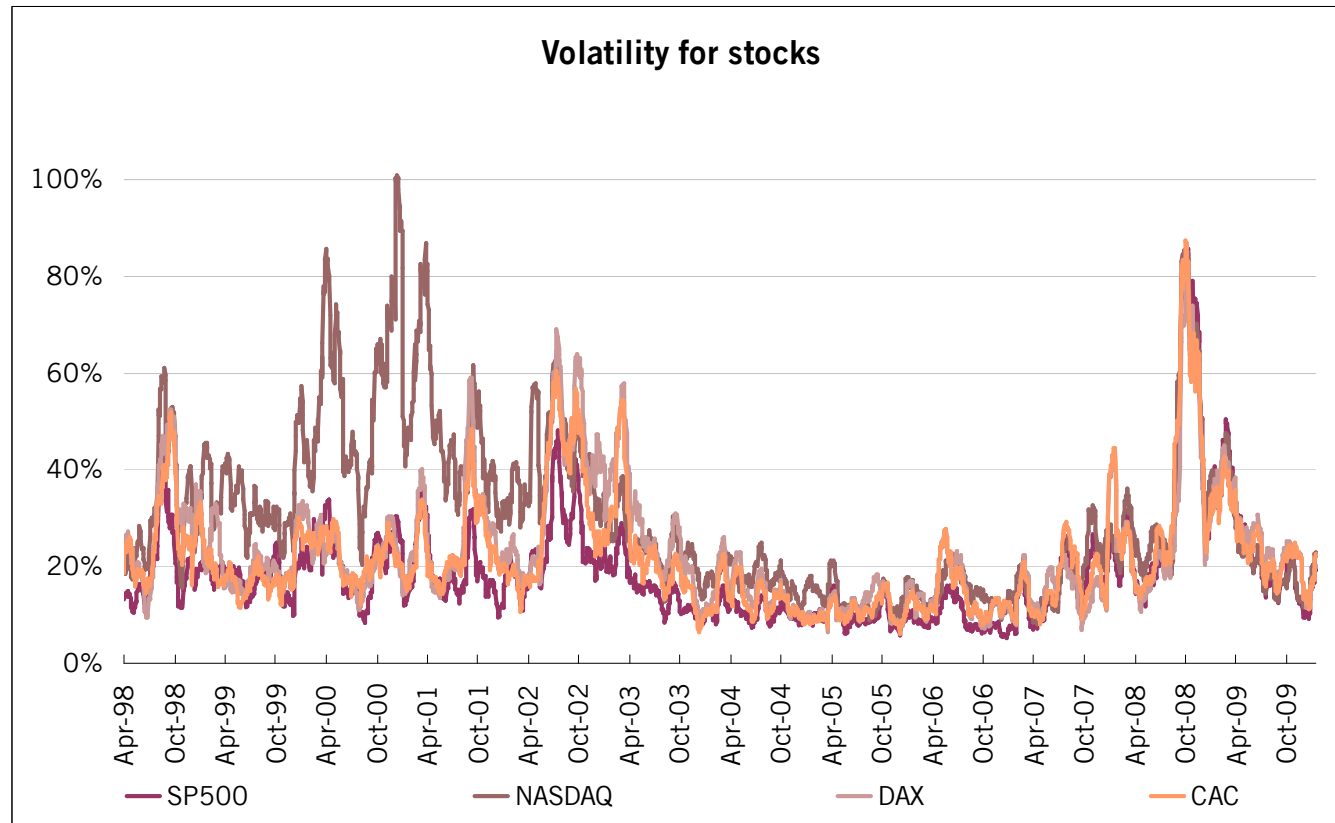
It is related to different fundamentals:

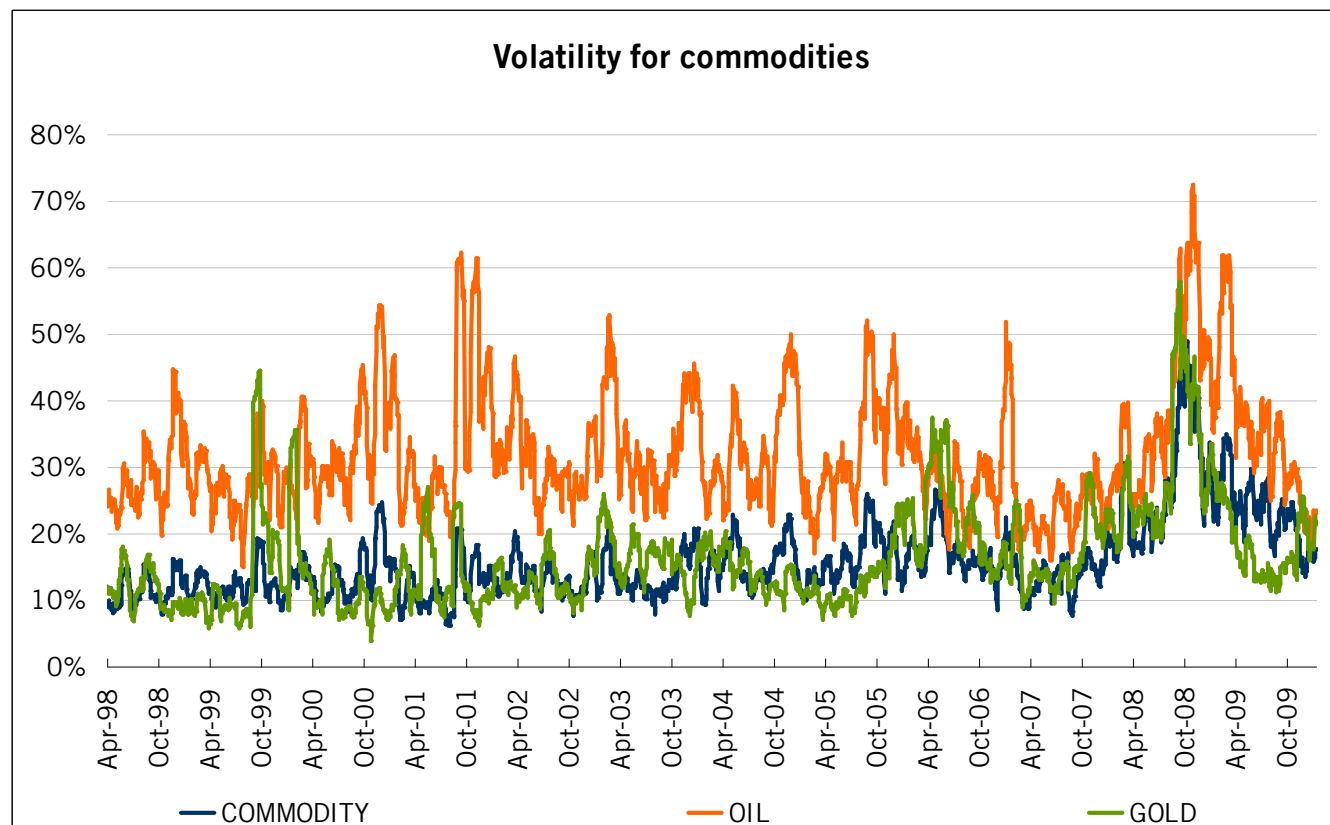
- Global economic stance of the world – developed and emerging markets: as profit perspectives drop, bankruptcy risks rise sharply and prices drop.

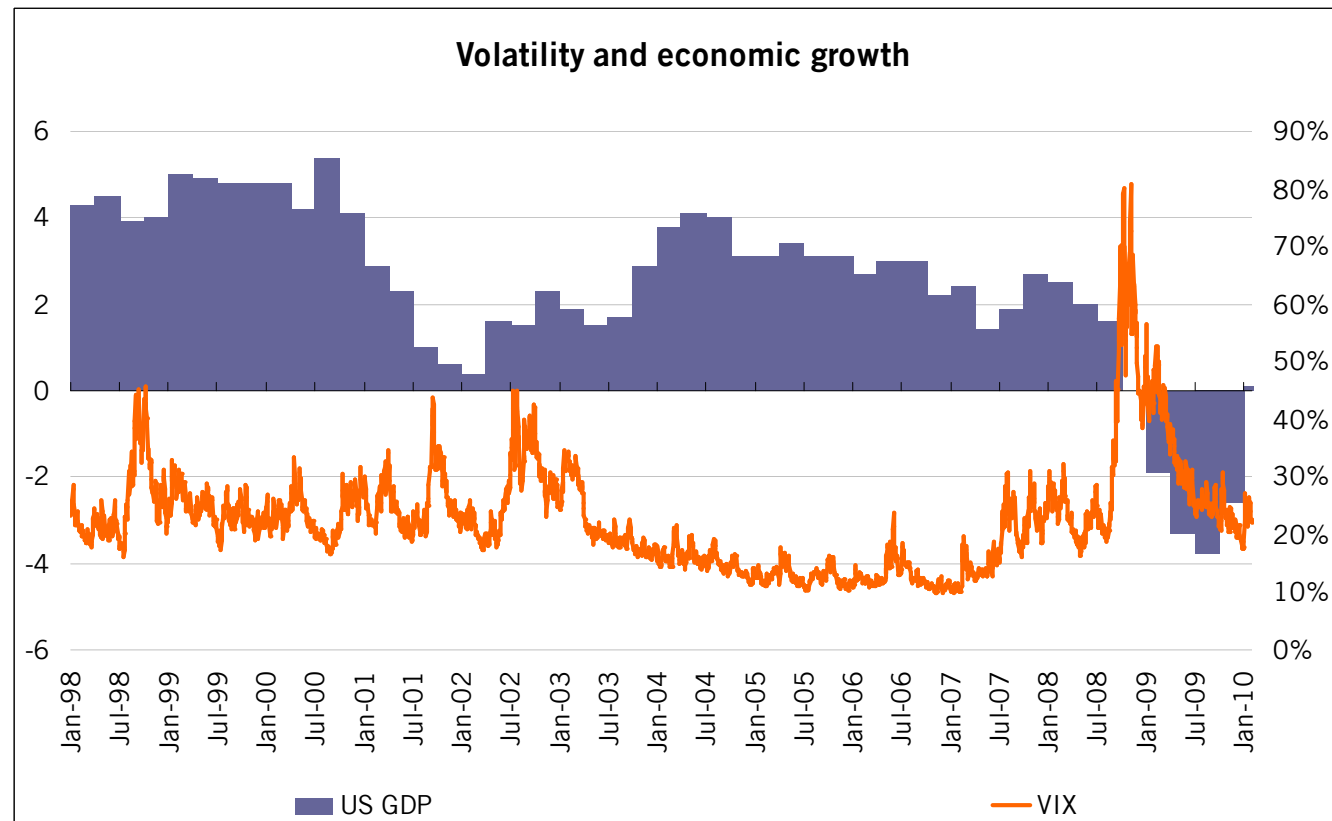
These periods are cyclical.

Volatility cycles match the NBER dating of economic crises.

- The order relation between asset classes is however globally unchanged: the volatility of low volatility asset classes remain below the one of high volatility asset classes.
- Most of the asset classes have a volatility that move together.







Stylized fact #2: the clustering effect

The key elements:

1. Volatility happens by clusters.
2. Periods of high volatility have a tendency to last for periods of a lower length than low volatility ones.

Rationale:

- Volatility reflects fear amongst of market participant. When the market is shocked, it takes some time for the shock to vanish.
- Volatility is related to the business cycle: crisis last for several months. Periods of expansion are usually longer than periods of expansion.
- Volatility reflects fear toward specific cases. In the 2010 Greek debt case, volatility cannot go down before a institutional solution is proposed.

Stylized fact #2: the clustering effect

In a few textbooks, volatility clustering is disentangled from the Taylor (1986) effect.

Absolute values for returns is the returns' transformation creating the biggest autocorrelation.

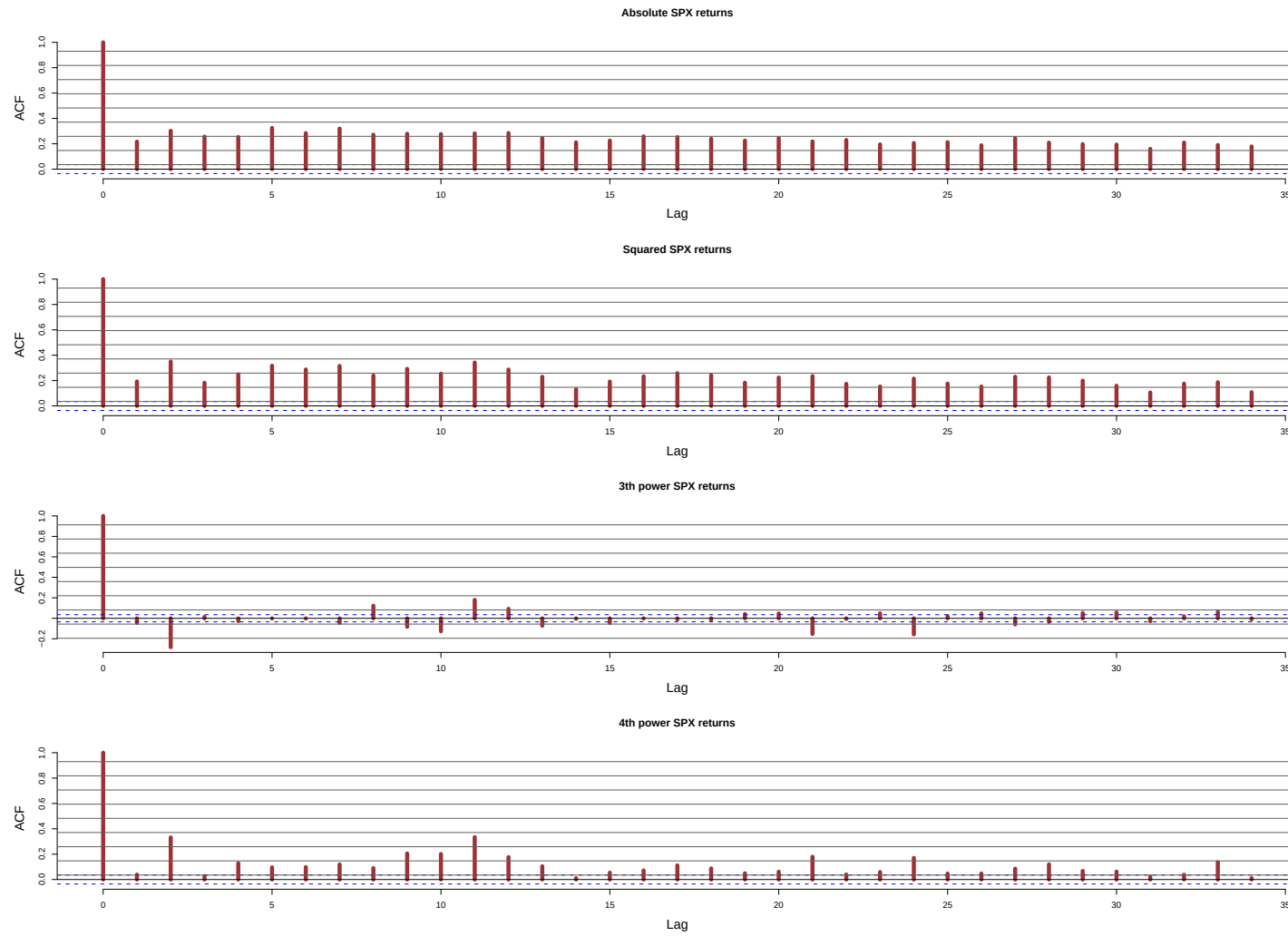
The autocorrelation function:

$$\gamma(h) = \frac{\text{Cov}(r_t, r_{t+h})}{\mathbb{V}[r_t]} \quad (7)$$

A process is said to be persistent whenever $\gamma(1)$ is statistically different from zero. The method of moment estimator has the following distribution:

$$\frac{\hat{\gamma}(h)}{\sigma(\hat{\gamma}(h))} \xrightarrow{\mathcal{L}} N(0, 1) \quad (8)$$

Rationale: Absolute returns are meant to represent volatility and volatility is a persistent process.



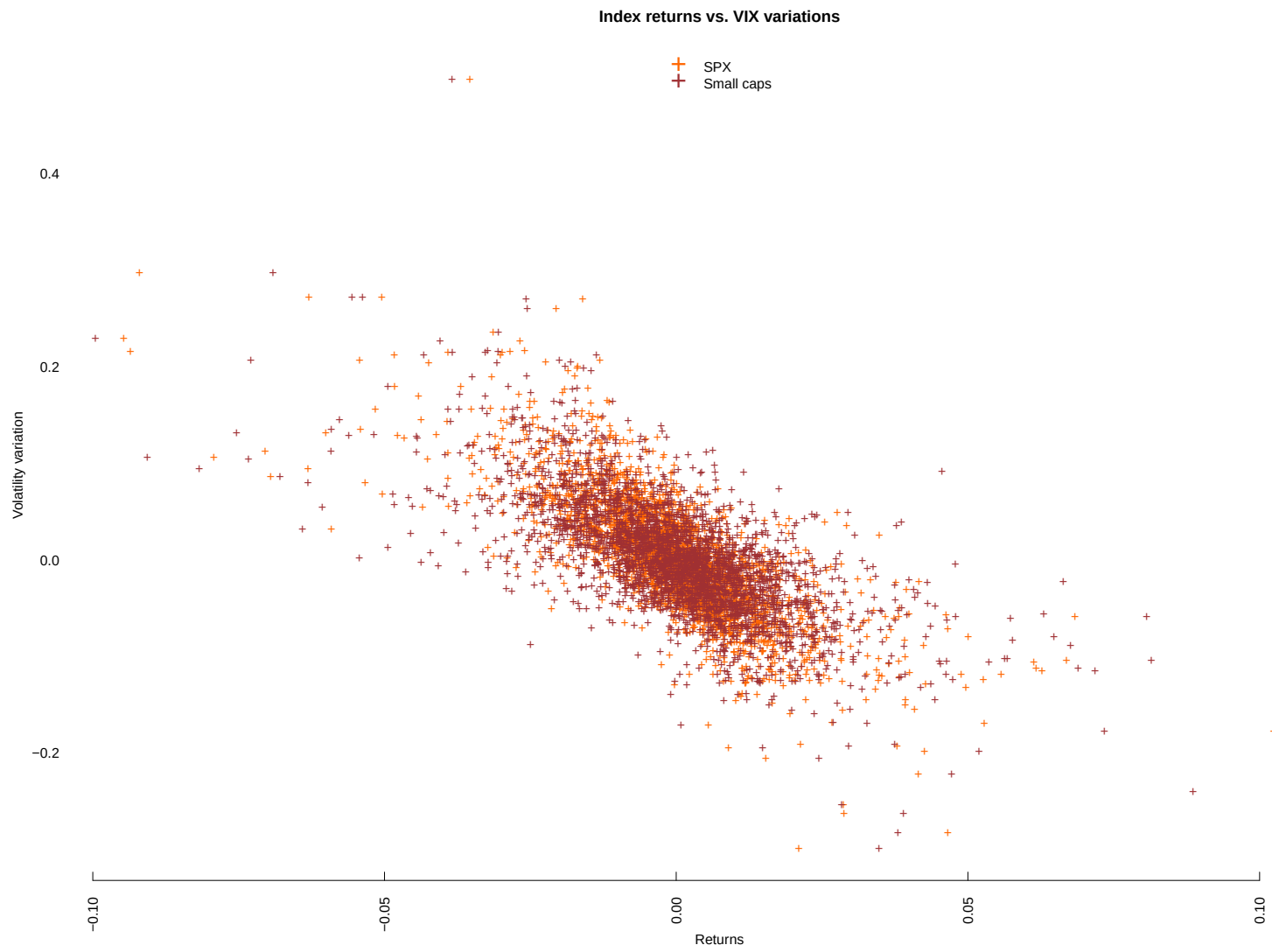
Stylized effect #3: the leverage effect

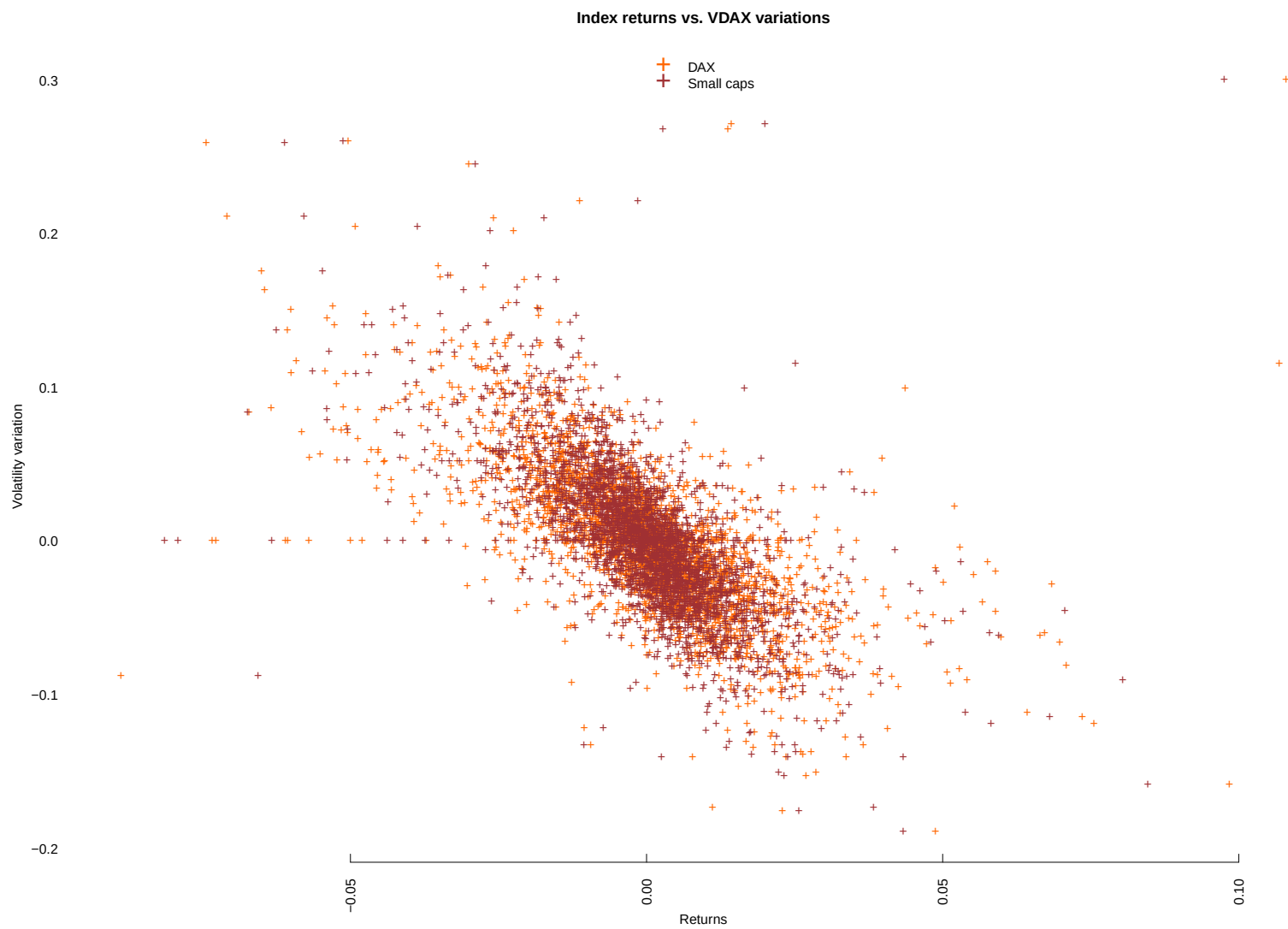
The key elements:

1. Volatility is higher during periods of negative returns.
2. Negative returns contribute more to a rise in volatility than positive ones.

Rationale:

- Volatility measures risk. Negative returns lead to a riskier world for investors. However, volatility does not measure only downward risks.
- For stocks, periods of economic expansion leads to an improving access to credit. The temptation to get leveraged (borrowing money to obtain several times the return on equity) is high, increasing in the mean time the economic consequences of a bankruptcy. Hence, the net loss perspectives are higher than the net gains.
- For commodities for which the supply is limited, the effect is converse: periods of high vol coincides with periods of positive returns, indicating an expected shortage. However, as in the stock case, long positions are more common than short ones.





Leverage effect

Correlation between returns and variations of VIX and VDAX

	VIX	VDAX
SP500	-76%	-39%
NASDAQ	-64%	-31%
DAX	-42%	-67%
CAC	-41%	-66%
SC.US	-68%	-38%
SC.EU	-42%	-68%
HY.EU	-1%	-2%
CORP.EU	6%	12%
CHINA	-11%	-20%
EM	-99%	-47%
EU.GOV	8%	14%
COMMODITY	-14%	-15%
OIL	-9%	-8%
GOLD	-2%	-3%
WORLD.GOV	18%	25%